

# SEULCHAN LEE

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[Google Scholar](#)

## Postdoctoral Researcher

Construction and Environmental Research Center

Earth and Remote Sensing Lab.

Sungkyunkwan University (SKKU)

2066 Seobu-ro, Suwon 16419, Republic of Korea

## EDUCATION

2019 – 2025 **Ph.D. in Water Resources**

Sungkyunkwan University, Suwon, Republic of Korea (advisor: [Dr. Minha Choi](#))

Dissertation: *Integrating Intermediate-Scale Observations and Land Surface Modeling for Synthetic Aperture Radar Soil Moisture Retrieval* [\[Link\]](#)

2019 **B.S. in Civil and Environmental Engineering**

Sungkyunkwan University, Suwon, Republic of Korea

## ACADEMIC & PROFESSIONAL EXPERIENCE

2025 – now **Postdoctoral Researcher** (advisor: [Dr. Minha Choi](#))

Construction and Environmental Research Center, Sungkyunkwan University, Suwon, Republic of Korea

2026 **Lecturer**

Dept. of Water Resources, Sungkyunkwan University, Suwon, Republic of Korea:

- (GWR5048) *Statistical Hydrology*, Spring Sems. 2026

2019 – 2025 **Graduate Research Assistant**

Dept. of Water Resources, Sungkyunkwan University, Suwon, Republic of Korea

2019 – 2024 **Teaching Assistant**

School of Civil, Architectural Eng. & Landscape Architecture, Dept. of Water Resources, Dept. of Civil and Environmental Engineering, Sungkyunkwan University:

- (CAL3086) *Remote Sensing of Global Environment*, Fall Sems. 2025
- (CAL3086) *Remote Sensing of Global Environment*, Fall Sems. 2024
- (GWR5047) *Water Resources & Remote Sensing*, Spring Sems. 2020
- (GWR5048) *Statistical Hydrology*, Fall Sems. 2019
- (CAL2015) *Introduction to Environmental Engineering*, Fall Sems. 2019
- (CAL3037) *Hydrology*, Spring Sems. 2019
- (CAL2014) *Fundamentals of Environmental Engineering and Science*, Spring Sems. 2019

## RESEARCH INTERESTS

- Water, Ecosystem and Environmental Science; Land surface-atmosphere interactions; Remote sensing
- Soil moisture dynamics; Evapotranspiration; Rainfall-Runoff simulation; Land surface modeling
- Flood and Drought analysis; Data Assimilation

## HONORS & AWARDS

2024 Dean's List, Sungkyunkwan University (1<sup>st</sup> semester)

2019 – 2021 Outstanding Graduate Student Scholarship, Sungkyunkwan University

2019 – 2020 Teaching and Research Assistant Scholarship, Sungkyunkwan University

2018 Dean's List, Sungkyunkwan University (1<sup>st</sup>, 2<sup>nd</sup> semester)

2017 Dean's List, Sungkyunkwan University (2<sup>nd</sup> semester)

2013 – 2018 National Science & Technology Scholarship, Korea Student Aid Foundation (KOSAF)

## PUBLICATIONS

### PEER-REVIEWED JOURNAL ARTICLES (SCI)

- 2026 D. Kim, **S. Lee**, and M. Choi\*. 2026. Global CMIP6-based datasets of vegetation-responsive and feedback-deflated potential evapotranspiration. *Scientific Data*. In production (Jun. 2026).  
<https://doi.org/10.1038/s41597-026-07553-0>
- 2026 **S. Lee**, S. Cho, and M. Choi\*. 2026. A novel approach for soil moisture retrieval from Sentinel-1 SAR via temporal stability-based backscatter analysis. *Remote Sensing of Environment*. 337, 115319.  
<https://doi.org/10.1016/j.rse.2026.115319>
- 2025 D. Kim, **S. Lee**, S. Cho, D. Kim, and M. Choi\*. 2025. Evaluating rainfall estimates derived from soil moisture using soil hydraulic properties over the Korean Peninsula. *Journal of Hydrology*. 663B, 134267.  
<https://doi.org/10.1016/j.jhydrol.2025.134267>
- 2025 **S. Lee**, J. Park, J. Jeong, and M. Choi\*. 2025. Soil moisture and evapotranspiration responses to precipitation uncertainty using Noah-MP land surface model. *Stochastic Environmental Research and Risk Assessment*. 2025, 1-20. <https://doi.org/10.1007/s00477-025-02991-5>
- 2024 W. Kim, **S. Lee**, and M. Choi\*. 2024. A comparative study of remotely sensed reservoir monitoring across multiple land cover types. *Science of the Total Environment*. 948, 174678.  
<https://doi.org/10.1016/j.scitotenv.2024.174678>
- 2024 J. Jeong, **S. Lee**, S. Cho, K. Kim, and M. Choi\*. 2024. Multi-scale assessment of a cosmic-ray neutron probe observation of soil moisture for surface layer applications in a mountainous forest environment. *Agricultural and Forest Meteorology*. 356, 110155. <https://doi.org/10.1016/j.agrformet.2024.110155>
- 2023 **S. Lee**, S. Oh, R.L. Ray, Y. Lee, and M. Choi\*. 2023. Three-dimensional hydrological thresholds to predict shallow landslides. *Terrestrial, Atmospheric and Oceanic Sciences*. 34, 20(2023).  
<https://doi.org/10.1007/s44195-023-00052-4>
- 2021 J. Jeong, **S. Lee**, and M. Choi\*. 2021. Correction efficiency and error characteristics for cosmic-ray soil moisture on mountainous terrain. *Journal of Hydrology*. 601, 126657.  
<https://doi.org/10.1016/j.jhydrol.2021.126657>

### ONGOING WORKS & UNDER REVIEW

- 2026 M. Kang, **S. Lee (co-first)**, D. Kim, and M. Choi\*. 2026. Hydroclimatic Controls on Flash Drought Development in Northeast Asia: Roles of Atmospheric Demand and Water Availability. *Hydrological Processes*. **Major Revision submitted** (Apr. 2026).
- 2026 S. Cho, **S. Lee (co-first)**, D. Kim, and M. Choi\*. 2026. Soil moisture retrieval from Sentinel-1 SAR using a water cloud model-constrained physics-informed neural network. *GIScience & Remote Sensing*. **Submitted** (Jun. 2026).
- 2026 W. Kim, S. Cho, **S. Lee**, H. Jeon, J.M. Jacobs, and M. Choi\*. 2026. Extraction of Water Bodies from SAR Imagery and Swin Transformer Model in Diverse Hydrological Conditions. *In Preparation*.
- 2026 **S. Lee**, D. Kim, K. Kim, and M. Choi\*. 2026. CRNP as the Regional Truth: Assessing the Representativeness of In-Situ Sensors for CRNP Calibration in Heterogeneous Terrains. *In Preparation*.

### PEER-REVIEWED KOREAN JOURNAL ARTICLES (KCI)

- 2024 S. Lee, W. Kim, **S. Lee**, H. Jeong, J. Park, and M. Choi\*. 2024. Comparative study of flood detection methodologies using Sentinel-1 satellite imagery. *Journal of Korea Water Resources Association*. 57(3), 181-193. <https://doi.org/10.3741/JKWRA.2024.57.3.181>
- 2023 M. Kang, J. Jeong, **S. Lee**, and M. Choi\*. 2023. A Study on Spatial Onset Characteristics of Flash Drought based on GLDAS Evaporative Stress in the Korean Peninsula. *Journal of Korea Water Resources Association*. 56(10), 631-639. <https://doi.org/10.3741/JKWRA.2023.56.10.631>
- 2023 H. Jeon, **S. Lee**, Y. Lee, J. Kim, and M. Choi\*. 2023. Evaluation of Bias and Uncertainty in Snow Depth Reanalysis data over South Korea. *Journal of Korea Water Resources Association*. 56(9), 543-551.  
<https://doi.org/10.3741/JKWRA.2023.56.9.543>
- 2023 **S. Lee**, W. Kim, S. Cho, H. Jeon, and M. Choi\*. 2023. Water Resources Monitoring Technique using Multi-

- source Satellite Image Data Fusion. *Journal of Korea Water Resources Association*. 56(8), 497-508. <https://doi.org/10.3741/jkwra.2023.56.8.497>
- 2023 J. Jeong, S. Cho, **S. Lee**, K. Kim, Y. Lee, C.D. Lee, S. Lee, and M. Choi\*. 2023. A Study on Domestic Applicability for the Korean Cosmic-Ray Soil Moisture Observing System. *Korean Journal of Remote Sensing*. 39(2), 233-246. <https://doi.org/10.7780/kjrs.2023.39.2.9>
- 2022 **S. Lee**, J. Jeong, S. Oh, H. Jeong, and M. Choi\*. 2022. Multi-resolution SAR Image-based Agricultural Reservoir Monitoring. *Korean Journal of Remote Sensing*. 38(5), 497-510. <https://doi.org/10.7780/kjrs.2022.38.5.1.5>
- 2022 J. Jeong, S. Cho, H. Jeon, **S. Lee**, and M. Choi\*. 2022. Assessment of Stand-alone Utilization of Sentinel-1 SAR for High Resolution Soil Moisture Retrieval Based on Machine Learning. *Korean Journal of Remote Sensing*. 38(5), 571-585. <https://doi.org/10.7780/kjrs.2022.38.5.1.11>
- 2022 S. Choi, Y. Youn, J. Kang, G. Park, G. Kim, **S. Lee**, M. Choi, H. Jeong, and Y. Lee\*. 2022. An Artificial Intelligence Approach to Waterbody Detection of the Agricultural Reservoirs in South Korea Using Sentinel-1 SAR Images. *Korean Journal of Remote Sensing*. 38(5), 925-938. <https://doi.org/10.7780/kjrs.2022.38.5.3.10>
- 2022 K. Kim, **S. Lee**, Y. Lee, M. Yeon, G. Lee, and M. Choi\*. 2022. Surface soil moisture memory using stored precipitation fraction in the Korean peninsula. *Journal of Korea Water Resources Association*. 55(2), 111-120. <https://doi.org/10.3741/jkwra.2022.55.2.111>
- 2021 J. Jeong, S. Oh, **S. Lee**, J. Kim, and M. Choi\*. 2021. Sentinel-1 SAR image-based waterbody detection technique for estimating the water storage in agricultural reservoirs. *Journal of Korea Water Resources Association*. 54(7), 535-544. <https://doi.org/10.3741/jkwra.2021.54.7.535>
- 2021 S. Cho, J. Jeong, **S. Lee**, and M. Choi\*. 2021. Estimation of soil moisture based on Sentinel-1 SAR data: Assessment of soil moisture estimation in different vegetation condition. *Journal of Korea Water Resources Association*. 54(2), 81-91. <https://doi.org/10.3741/jkwra.2021.54.2.81>
- 2020 S. Oh, J. Jeong, **S. Lee**, and M. Choi\*. 2020. Downscaling GPM Precipitation Using Finer-scale MODIS Based Optical Image in Korean Peninsula. *Korean Journal of Remote Sensing*. 36(5), 749-762. <https://doi.org/10.7780/kjrs.2020.36.5.1.8>
- 2020 S. Cho, J. Jeong, **S. Lee**, and M. Choi\*. 2020. Estimation of soil moisture based on sentinel-1 SAR data: focusing on cropland and grassland area. *Journal of Korea Water Resources Association*. 53(11), 973-983. <https://doi.org/10.3741/jkwra.2020.53.11.973>
- 2020 H. Jeon, J. Baik, **S. Lee**, and M. Choi\*. 2020. A point-scale gap filling of the flux-tower data using the artificial neural network. *Journal of Korea Water Resources Association*. 53(11), 929-938. <https://doi.org/10.3741/jkwra.2020.53.11.929>
- 2019 **S. Lee**, J. Baik, M. Choi, and Y. Cho\*. 2019. Evaluation of the behavior and quality in soil moisture data: A case study of Yongdam study watershed. *Journal of Korea Water Resources Association*. 52(12), 951-962. <https://doi.org/10.3741/jkwra.2019.52.12.951>
- 2019 **S. Lee**, J. Jeong, J. Park, H. Jeon, and M. Choi\*. 2019. Estimating Fine Particulate Matter Concentration using GLDAS Hydrometeorological Data. *Korean Journal of Remote Sensing*. 35(6-1), 919-932. <https://doi.org/10.7780/kjrs.2019.35.6.1.4>
- 2019 J. Baik, S. Cho, **S. Lee**, and M. Choi\*. 2019. Analysis on Adequacy of the Satellite Soil Moisture Data (AMSR2, ASCAT, and ESACCI) in Korean Peninsula: With Classification of Freezing and Melting Periods. *Korean Journal of Remote Sensing*. 35(5-1), 625-636. <https://doi.org/10.7780/kjrs.2019.35.5.1.1>

## ARTICLES IN PROFESSIONAL MAGAZINES

- 2026 **S. Lee** and M. Choi. 2026. Overcoming the spatial representativeness limitations in remote sensing hydrology: validation of high-resolution SAR soil moisture retrievals using cosmic-ray neutron sensing (CRNS). *Water for Future* (Korea Water Resources Association).
- 2022 **S. Lee**, M. Kang, and M. Choi. 2022. Reservoir water body monitoring using multi-satellite images. *Water for Future* (Korea Water Resources Association). 55(12), 6-10. [\[Link\]](#)
- 2022 S. Oh, **S. Lee**, W. Kim, and M. Choi. 2022. Water body detection technique using Sentinel-1 SAR images. *Water for Future* (Korea Water Resources Association). 55(6), 47-51. [\[Link\]](#)
- 2021 J. Jeong, S. Cho, **S. Lee**, S. Oh, H. Jeon, W. Kim, and M. Choi. 2021. Artificial Intelligence, a future partner

for hydrological analyses. *Water for Future* (Korea Water Resources Association). 54(12), 8-14. [\[Link\]](#)

## CONFERENCE PROCEEDINGS, POSTERS, AND PRESENTATIONS

### INTERNATIONAL CONFERENCES

- 2026 D. Kim, **S. Lee**, J. Jeong, S. Cho, and M. Choi. 2026. Exploring Three-dimensional Soil Moisture Variability using Multi-Source Observations. *EGU General Assembly 2026*, 2026.05.03-08, Vienna, Austria
- 2026 J. Jeong, D. Kim, **S. Lee**, W. Kim, and M. Choi. 2026. Integration of SMAP L-band VOD and Multi-source Satellite Data for Improved Sentinel-1 Soil Moisture Retrieval in Complex Terrains. *EGU General Assembly 2026*, 2026.05.03-08, Vienna, Austria
- 2025 D. Kim, S. Cho, **S. Lee**, and M. Choi. 2025. Spatiotemporal Variability of Soil Moisture: Multi-Sensor Observations at the Eoseom Korea Hydrometeorological Observatory. *AGU 2025 Fall Meeting*, 2025.12.15-19, Online
- 2025 J. Jeong, **S. Lee**, D. Kim, and M. Choi. 2025. High-Resolution Root Zone Soil Moisture Modeling for Small Watersheds using the Land Information System and Multi-Product Comparison. *2025 KAGIS Autumn Conference & IAG'i International Symposium*, 2025.11.06-08, Jeju, Republic of Korea
- 2025 **S. Lee**, J. Jeong, and M. Choi. 2025. Improving Soil Moisture Retrieval Accuracy with Temporal Stability-based Masking of SAR backscatter. *SPIE Sensors + Imaging 2025*, 2025.09.15-18, Madrid, Spain
- 2025 H. Jeon, **S. Lee**, and M. Choi. 2025. High-Resolution Evapotranspiration Modeling from Satellite and Drone Observations via METRIC. *SPIE Sensors + Imaging 2025*, 2025.09.15-18, Madrid, Spain
- 2025 J. Jeong, **S. Lee**, D. Kim, and M. Choi. 2025. Prediction and Evaluation of Hydrological Factors and Drought Indices under Climate Change using CMIP6 in Monsoon Regions. *EGU General Assembly 2025*, 2025.04.27-05.02, Vienna, Austria
- 2024 M. Kang, **S. Lee**, and M. Choi. 2024. Role of Terrestrial Evapotranspiration on the Occurrence of Flash Droughts in Northeast Asia. *IGARSS 2024*, 2024.07.07-12, Athens, Greece
- 2024 S. Cho, **S. Lee**, H. Jeon, and M. Choi. 2024. SAR Soil Moisture Retrieval using Physically based AI Model. *IGARSS 2024*, 2024.07.07-12, Athens, Greece
- 2024 **S. Lee**, J. Park, and M. Choi. 2024. Precipitation Forcing-induced Uncertainties in Land Surface Model (LSM) Simulations. *AOGS 2024 21st Annual Meeting*, 2024.06.23-28, Pyeongchang, Republic of Korea
- 2024 H. Jeon, **S. Lee**, and M. Choi. 2024. Evaluation of LDAPS Snow information by MERRA-2 and ASOS over the South Korea. *EGU General Assembly 2024*, 2024.04.14-19, Vienna, Austria
- 2023 **S. Lee**, J. Jeong, N.S.M. Harmay, and M. Choi. 2023. Impacts of Precipitation Forcing on Hydrological Simulation over Monsoon Asia. *EGU General Assembly 2023*, 2023.04.23-28, Vienna, Austria
- 2023 W. Kim, **S. Lee**, and M. Choi. 2023. Flooded area monitoring using SAR image-based water body detection technique. *EGU General Assembly 2023*, 2023.04.23-28, Vienna, Austria
- 2023 J. Jeong, S. Cho, **S. Lee**, K. Kim, C.D. Lee, S. Lee, and M. Choi. 2023. Evaluation of Cosmic-Ray Neutron Probe Based Soil Moisture Observatory for Korean cOsmic-ray Soil Moisture Observing System (KOSMOS). *The 28th International Symposium on Remote Sensing*, 2023.04.19-21, Jeju, Republic of Korea
- 2023 S. Cho, J. Jeong, **S. Lee**, and M. Choi. 2023. Soil Moisture Estimation and Mapping using Sentinel-1 on vegetated area in Korean Peninsula using Physical methods. *The 28th International Symposium on Remote Sensing*, 2023.04.19-21, Jeju, Republic of Korea
- 2023 W. Kim, **S. Lee**, and M. Choi. 2023. Agricultural reservoir monitoring using multi-dimensional satellite images. *The 28th International Symposium on Remote Sensing*, 2023.04.19-21, Jeju, Republic of Korea
- 2022 **S. Lee**, J. Park, J. Jeong, and M. Choi. 2022. Influence of Precipitation Forcing in Hydrological Simulation over Monsoon Asia. *AGU 2022 Fall Meeting*, 2022.12.12-16, Online
- 2022 S. Cho, **S. Lee**, S. Oh, and M. Choi. 2022. High resolution soil moisture estimation considering complex vegetated areas with modified Water Cloud Model. *AGU 2022 Fall Meeting*, 2022.12.12-16, Online
- 2022 S. Oh, **S. Lee**, W. Kim, and M. Choi. 2022. The use of Synthetic Aperture Radar for surface waterbody detection: a comparative study of thresholding methods. *AGU 2022 Fall Meeting*, 2022.12.12-16, Online
- 2021 M. Choi, **S. Lee**, J. Jeong, and S. Cho. 2021. Conditional Gap-filling of in-situ Soil Moisture for Continuous Water Monitoring. *IWRA World Water Congress*, 2021.11.29-12.03, Daegu, Republic of Korea

- 2020 M. Choi, H.H. Nguyen, J. Jeong, **S. Lee**, S. Oh, and S. Cho. 2020. Assessment of the soil moisture-evaporative fraction coupling strength using the Cosmic-Ray Neutron Sensing network. *AGU 2020 Fall Meeting*, 2020.12.07-11, Online
- 2020 **S. Lee**, H. Jeon, J. Park, and M. Choi. 2020. Performance of a Physically Based Gap-Filling Technique of in-situ Soil Moisture, in Comparison with Machine Learning. *EGU General Assembly 2020*, 2020.05.04-08, Online
- 2019 S. Cho, M. Zohaib, **S. Lee**, and M. Choi. 2019. Quantification of Irrigation water consumption using remote sensing dataset in Korea Peninsula. *AGU 2019 Fall Meeting*, 2019.12.09-13, San Francisco, U.S.
- 2019 **S. Lee**, J. Jeong, and M. Choi. 2019. Statistical Assessment of the Influence of Hydrometeorological factors on Fine Dust. *Asian Conference on Remote Sensing 2019*, 2019.10.14-18, Daejeon, Republic of Korea
- 2019 **S. Lee**, J. Jeong, and M. Choi. 2019. Hydrometeorological drivers of particulate matter using Bayesian model averaging. *IGARSS 2019*, 2019.07.28-08.02, Yokohama, Japan
- 2018 J. Baik, J. Park, **S. Lee**, U. Kim, and M. Choi. 2018. Assessment of merging technique using Triple Collocation (TC) from satellite and reanalysis dataset over Different Land Covers in East Asia: GLDAS, MOD16, GLEAM, and MERRA. *AGU 2018 Fall Meeting*, 2018.12.10-15, Washington D.C., U.S.

### KOREAN CONFERENCES

- 2026 J. Jeong, D. Kim, **S. Lee**, and M. Choi. 2026. Application of SMAP L-band VOD as a Vegetation Indicator in the Water Cloud Model for Sentinel-1 Soil Moisture Estimation over Complex Terrain. *The Korea Water Resources Association Conference*, 2026.05.20-22, Busan
- 2025 **S. Lee** and M. Choi. 2025. Temporal Stability-informed Backscatter Masking Strategy for Sentinel-1 SAR Soil Moisture Retrieval. *The Korean Society of Remote Sensing Conference*, 2025.10.22-24, Yeosu
- 2025 J. Jeong, **S. Lee**, D. Kim, and M. Choi. 2025. Calibration of CRNP Sensor Using Noah-MP LSM-simulated Soil Moisture Data. *The Korean Society of Remote Sensing Conference*, 2025.10.22-24, Yeosu
- 2025 **S. Lee**, S. Cho, and M. Choi. 2025. Enhancing high-resolution soil moisture retrieval from Sentinel-1 SAR: A temporal stability-based approach. *The Korea Water Resources Association Conference*, 2025.05.21-23, Yeosu
- 2025 S. Cho, **S. Lee**, and M. Choi. 2025. The SAR based regional scale soil moisture retrieval and evaluation using land surface information with Physics Informed Neural Network model. *The Korea Water Resources Association Conference*, 2025.05.21-23, Yeosu
- 2025 J. Jeong, **S. Lee**, and M. Choi. 2025. Projection of future drought indices through validation and bias correction of hydrological variables in East Asia using CMIP6. *The Korea Water Resources Association Conference*, 2025.05.21-23, Yeosu
- 2025 M. Choi, **S. Lee**, S. Cho, J. Yi, and S.J. Oh. 2025. A Study on SAR Soil Moisture Retrieval using Physics-based AI model. *Korean Society of Hazard Mitigation Conference*, 2025.02.19-21, Jeju
- 2024 W. Kim, **S. Lee**, and M. Choi. 2024. Multi-satellite remote sensing-based agricultural reservoir monitoring. *The Korean Society of Remote Sensing Conference*, 2024.10.16-18, Seoul
- 2024 M. Choi, W. Kim, and **S. Lee**. 2024. Remotely sensed agricultural reservoir monitoring in Korean peninsula. *The Korean Society of Civil Engineers Conference*, 2024.10.16-18, Jeju
- 2023 **S. Lee**, W. Kim, S. Lee, and M. Choi. 2023. Reservoir water body detection using multi-satellite remote sensing data in Korea. *The Korean Society of Remote Sensing Conference*, 2023.11.15-17, Gyeongju
- 2023 D. Kim, J. Jeong, H. Jeon, **S. Lee**, and M. Choi. 2023. Improving soil moisture accuracy in ungauged areas using Multi-Satellite data. *The Korea Water Resources Association Conference*, 2023.05.25-26, Gosung
- 2022 M. Kang, S. Cho, **S. Lee**, and M. Choi. 2022. Flash drought and its impact on eco-hydrology using MODIS over the Korean peninsula. *The Korean Association of Geographic Information Studies Conference*, 2022.11.03-05, Jeju
- 2022 M. Choi, **S. Lee**, W. Kim, and S. Oh. 2022. Comparative analysis of thresholding techniques for SAR based waterbody detection. *The Korean Society of Remote Sensing Conference*, 2022.11.07-09, Busan
- 2021 N.S.M. Harmay, **S. Lee**, and M. Choi. 2021. Characteristics of the Land Surface Variables at Multiple Land Cover Types under Summer Heatwaves Influences. *The Korean Society of Remote Sensing Conference*, 2021.10.20-22, Online

- 2021 W. Kim, **S. Lee**, J. Park, S. Fred, and M. Choi. 2021. Hydrological drought analysis using SAR image. *The Korean Society of Remote Sensing Conference*, 2021.10.20-22, Online
- 2021 M. Yeon, L.N. Van, **S. Lee**, and G. Lee. 2021. Analysis of soil erosion in agricultural lands using physics-based erosion model. *The Korea Water Resources Association Conference*, 2021.06.03-04, Gwangju
- 2021 K. Kim, **S. Lee**, M. Choi, S. Jung, and M. Yeon. 2021. Evaluation of multiple-satellite precipitation data by rainfall intensity. *The Korea Water Resources Association Conference*, 2021.06.03-04, Gwangju
- 2020 S. Cho, J. Jeong, **S. Lee**, and M. Choi. 2020. Estimation and mapping of high resolution soil moisture using Sentinel-1 SAR images. *The Korean Society of Remote Sensing Conference*, 2020.11.04-06, Gangchon
- 2019 J. Jeong, **S. Lee**, S. Cho, and M. Choi. 2019. An Analysis on the Relation between Particulate Matter and Hydrometeorological Factors Using Bayesian Model Averaging. *The Korean Society of Civil Engineers Conference*, 2019.10.16-18, Pyeongchang
- 2019 **S. Lee**, J. Jeong, and M. Choi. 2019. Hydrometeorological Drivers of Particulate Matter Using Satellite and Reanalysis Data. *The Korea Water Resources Association Conference*, 2019.05.30-31, Yeosu
- 2019 **S. Lee**, J. Jeong, and M. Choi. 2019. A Study on the Hydrometeorological Drivers of Particulate Matter. *The Korean Meteorological Society Conference*, 2019.04.22-23, Daegu
- 2018 **S. Lee**, J. Jeong, and M. Choi. 2018. A Study on the Relationship between Precipitation using GPM and Fine Dust Concentration in Spring Season. *The Korean Society of Remote Sensing Conference*, 2018.10.24-26, Muju

## RESEARCH PROJECT EXPERIENCES

- 2025 – now Net-zero strategy for sustainable water conservation and nature-based solutions, National Research Foundation of Korea (NRF), Postdoctoral Researcher
- 2024 – 2025 Net-zero strategy for sustainable water conservation and nature-based solutions, National Research Foundation of Korea (NRF), Research Assistant
- 2024 – 2025 Improving storage capacity of underground infrastructure in response to urban flooding and developing multi-functional utilization technology, Korea Environmental Industry and Technology Institute (KEITI), Research Assistant
- 2024 – 2025 Cycle adaptive coastal erosion management technology development, Korea Institute of Marine Science & Technology Promotion (KIMST), Research Assistant
- 2024 Research on the time series analysis techniques for national water surface area using medium-resolution SAR satellite imagery, National Disaster Management Research Institute (NDMI), Research Assistant
- 2024 – 2025 Development of AI-based water resources big data quality management technology, Korea Environmental Industry and Technology Institute (KEITI), Research Assistant
- 2023 – 2025 Development of damage analysis technology of water shortage and socio-economic value evaluation technology of water supply, and its application strategy, Korea Environmental Industry and Technology Institute (KEITI), Research Assistant
- 2023 Development of technology for nationwide reservoir surface area analysis and visualization using satellite data, National Disaster Management Research Institute (NDMI), Research Assistant
- 2022 – 2025 Development of comprehensive land management technology using satellite image information bigdata, Korea Agency for Infrastructure Technology Advancement (KAIA), Research Assistant
- 2022 – 2025 Highly precise 3-Dimensional spatial soil moisture tracing algorithm using artificial intelligence for short-term flood and drought prediction, National Research Foundation of Korea (NRF), Research Assistant
- 2022 – 2024 Development of big data-based disaster impact assessment technology for development projects, Ministry of the Interior and Safety (MOIS), Research Assistant
- 2022 Development of improved waterbody extraction technology using satellite radar data, National Disaster Management Research Institute (NDMI), Research Assistant
- 2021 – 2025 Development of intelligent disaster observation and surveillance technology using satellite imagery, Ministry of the Interior and Safety (MOIS), Research Assistant
- 2021 – 2022 Development of a physics-based modeling system for analysis and prediction of high-risk surface soil failure area, Korea Environmental Industry and Technology Institute (KEITI), Research Assistant

- 2021 Research on water management strategies based on national satellite imagery (Supporting the implementation of the Korea-Uzbekistan governmental space cooperation MoU), Korea Aerospace Research Institute (KARI), Research Assistant
- 2021 Development of the multi-satellite information supply and demand system for disaster safety management, National Disaster Management Research Institute (NDMI), Research Assistant
- 2019 – 2022 Blending of multinational/multispectral satellite-based soil moisture products for natural disaster forecast and analysis over Korean peninsula, National Research Foundation of Korea (NRF), Research Assistant
- 2018 – 2021 Development of the merged drought monitoring technology for natural disaster mitigation of satellite-based big data, National Research Foundation of Korea (NRF), Research Assistant
- 2018 – 2019 Development of integrated soil moisture measurement system for national water management, Korea Agency for Infrastructure Technology Advancement (KAIA), Research Assistant
- 2018 – 2019 Improving prediction of Asian dust events using COMS/Metop-A dataset, National Research Foundation of Korea (NRF), Research Assistant
- 2018 – 2019 Blending of multinational/multispectral satellite-based soil moisture products for natural disaster forecast and analysis over Korean peninsula, National Research Foundation of Korea (NRF), Research Assistant

## SOFTWARE & SYSTEM DEVELOPMENT

- 2021 Hydrometeorological data inventory system for satellite data hub, Supported by LINC+ Project, Sungkyunkwan University
- 2020 Development of satellite data sharing service, Supported by LINC+ Project, Sungkyunkwan University

## PROFESSIONAL AFFILIATIONS

- 2025 – now The Korean Association of Geographic Information Studies (KAGIS), member
- 2025 – now Society of Photo-Optical Instrumentation Engineers (SPIE), member
- 2024 – now Asia Oceania Geosciences Society (AOGS), member
- 2019 – now European Geosciences Union (EGU), member
- 2019 – now Korea Water Resources Association (KWRA), member
- 2019 – now Korean Meteorological Society (KMS), member
- 2019 – now Korean Society of Civil Engineers (KSCE), member
- 2018 – now American Geophysical Union (AGU), member
- 2018 – now The Korean Society of Remote Sensing (KSRS), member

## TECHNICAL SKILLS

### LANGUAGES

Proficient in Korean / English  
 Limited in Japanese

### HARDWARE, SOFTWARE, PROGRAMMING LANGUAGE

Matlab, Python, Interactive Data Language (IDL), ENVI, QGIS, R, SNAP  
 Land Information System Framework (LISF), Community Land Model (CLM)  
 Sensor techniques (Campbell Scientific CS616/CS625 soil moisture sensors, METER TEROS 12 soil moisture sensor, Hydroinnova CRS-1000 cosmic-ray sensor)

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## REFERENCES

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